

Dinesh Kumar Gadde

Buffalo, NY 14221 | +1 716-750-6215 | dineshgadden@gmail.com | linkedin.com/in/dinesh-kumar-gadde

SUMMARY

Motivated IoT and Mechatronics Engineer with 3+ years of hands-on experience in embedded systems, electronics testing, and industrial automation. Expertise in circuit design, PLC programming (SIEMENS S7-300), and IoT protocols (BLE, Zigbee, Z-Wave, LoRaWAN). Proven ability to develop, validate, and optimize hardware and firmware solutions with measurable impact on product quality and operational efficiency. Seeking challenging roles in IoT product development and embedded systems engineering.

TECHNICAL SKILLS

Core Competencies: Basic Electronics, Electrical Testing and Troubleshooting, Internet of Things (IoT), Embedded Systems, Robotics, Industrial Automation, Mechatronics, Electrical Engineering, Manufacturing, Test Equipment, Home Automation, PLC Programming (SIEMENS S7-300), Pneumatics, Hydraulics, Machine Maintenance

Programming Languages: Python, C, C++, HTML, CSS, Ladder Logic

Tools & Software: Anaconda Navigator, VS Code, Google Colab, GitHub, AWS, Arduino IDE, Keil, KiCad (PCB Design), AutoCAD, MATLAB, JTAG, Creo Parametric ProE, Autodesk Fusion 360, MS Office, Wireshark

Hobbies: Making Telugu Podcasts, Electronics Hobby Projects

EXPERIENCE

Associate Engineer

Hogar Controls India

Mar 2023 – Jun 2024

Hyderabad, India

Spearheaded physical testing of 10+ smart devices, improving quality and performance standards by 15%.

Developed and refined embedded firmware, designed mechanical components, and mapped complex I/O systems using BLE, Wi-Fi, Zigbee, Z-Wave protocols.

Collaborated with cross-functional teams on Android and iOS application testing and debugging for seamless integration.

Authored 50+ test cases and maintained detailed compliance reports supporting continuous improvement initiatives.

Utilized Google Workspace, Jira, Zoho, and Slack for documentation and project tracking.

Trainee Mechatronics Engineer

ZF Steering Gear India Pvt Ltd

Mar 2018 – Mar 2020

Pune, India

Managed and maintained PLC-controlled machines with pneumatics, servo motors, and sensors on fully automated production lines, consistently exceeding daily targets.

Ensured quality assurance, supervised production operations, and implemented 5S and Kaizen methodologies for process optimization.

Meticulously maintained production reports for data-driven decision-making and continuous improvement.

PROJECTS

Campus Connectivity on the Move: IoT Robotic Car (Wi-Fi vs 5G Study)

Master's Capstone Project

Aug 2025 – Dec 2025

Raspberry Pi, 5G LTE HAT, Python

Developed IoT robotic car testbed using Raspberry Pi and 5G LTE HAT for real-world network performance benchmarking.

Conducted field experiments measuring throughput, latency, jitter, and reliability across campus environments.

Delivered actionable recommendations for reliable, low-latency IoT network deployment in smart campus infrastructure.

Home Automation Using Bluetooth and MIT App Inventor

Mini IoT Course Project

Oct 2024

ESP32, DHT Sensor, Relay Modules, Blynk

Designed smart home automation system using ESP32, DHT sensor, and relay modules controlled via custom Blynk mobile app.

Implemented real-time temperature/humidity monitoring and remote switching for up to four appliances via Wi-Fi and Bluetooth.

Integrated error handling and cloud dashboard for live status updates and robust user experience.

Green Hati: Machine for Recycling Thermocol Plates

Jun 2020 – Sep 2020

Research Project

Sensors, Mechanical Design, Chemical Processes

Engineered automated system converting single-use thermocol plates to fuel pellets using sensor-driven mechanical and chemical processes.

Optimized design for efficient recycling, reducing landfill waste and enabling sustainable energy at large-scale events.

Published technical results in JETIR, validating scalability and real-world impact.

Electric Linear Actuator with Remote Control Shaft Adjustment

Oct 2019 – Feb 2020

Engineering Project

PCB Design, Mechanical Enclosure

Prototyped and designed PCB and mechanical enclosure for wire-remote control box supporting four actuator models with varying payloads and ± 2 cm adjustment accuracy.

Coordinated mechanical component manufacturing and led final assembly with integrated safety limit switches.

Delivered robust, customized automation solutions meeting rigorous specifications and ergonomic requirements.

Multiple Adjustment Wheelchair with Wireless Alert

Aug 2017 – Jun 2018

Diploma's Capstone Project (Published)

RF Transceiver, GSM, Microcontrollers

Developed smart wheelchair with RF transceiver-based remote control for backrest and footrest adjustments.

Integrated pulse rate sensor and LCD for live health monitoring.

Implemented GSM-based emergency alert system for automatic SMS notifications.

Assembled DC motor, relay, and microcontroller circuits for safe, precise operation.

EDUCATION

University at Buffalo, The State University of New York

Buffalo, NY

Master of Science, Engineering Science (Internet of Things)

Aug 2024 – Jan 2026

GPA: 3.87/4.0

Lovely Professional University

India

Bachelor of Technology in Mechatronics Engineering

June 2020 – June 2023

GPA: 3.4/4.0

PUBLICATIONS

Effect of hybridization on mechanical properties of unidirectional Sisal/Banana hybrid composites (2024) – DOI:10.1063/5.0192292

Multiple Adjustment Wheelchair with Wireless Alert (2022) – JETIR2211303

Published Review Paper on Waste Processing Machines (JETIR, 2021) – JETIREL06178

CERTIFICATIONS

NSDC Automation Specialist – Nettur Technical Training Foundation, Bangalore, 2018

Academy for LoRaWAN – Semtech, October 2025